THE VALUE, WITH ITS EVIDENCE.

# Sample engine report

## 01 Executive summary

3 years · USD · discount rate 10.0%

| Measure | Hard savings | Capacity-inclusive |
|---|---|---|
| Net present value | -$18,641.93 | $127,956.73 |
| ROI · undiscounted | -15.6% | 202.9% |
| ROI · NPV / PV of costs | -24.8% | 169.9% |
| Payback | No payback within the selected horizon | 1.0 years (uniform-flow assumption) |
| Benefits · present value | All included drivers | $203,275.29 |
| Costs · present value | All cost lines | $75,318.56 |

**Project confidence 93.5% · floor 85.0%**

7 documented inputs, 0 measured. Effective haircuts: time savings 6.5%, data quality 4.5%, risk reduction 6.5%. 0 measured · 7 documented · 0 benchmark · 0 estimate · 0 assumption. Capped by time_savings.

Over three years this project is estimated to return a capacity-inclusive NPV of $127,957 on costs with a present value of $75,319 [D1, D2, D3]. Hard savings alone give an NPV of $-18,642 [B3]. All inputs are documented rather than measured, so treat this as a pre-funding estimate.

**STUB · not a live assessment**

This report renders the supplied PR #123 sample. The real engine, saved intake, access checks and PDF storage are not connected. The exporter formats supplied values and does not recompute financial metrics.

## 02 Context and assumptions

Run run_sample_released. Expected scenario. Generated 2026-11-05T12:00:00Z. Contract 1; formula 0.1; settings s1.

**Input snapshot pending**

Step input arrays in this sample are empty. The source formulas and source types are preserved below; no missing entered inputs or project context have been invented.

| Accepted proxy | Value | Source / tier |
|---|---|---|
| recapture_pct | 50 | Nucleus productivity correction factor 0.5 · benchmark |
| discount_rate_pct | 10 | Forrester TEI convention (organizations typically 8-16%) · benchmark |

## 03 Benefits by driver

Adjusted values over the selected horizon

| Driver / class | Haircut | Confidence | Total | PV |
|---|---|---|---|---|
| Time savings · Capacity value · included | 6.5% | 93.5% | $102,850.00 | $83,770.66 |
| Data quality · Hard savings · included | 4.5% | 95.5% | $69,585.12 | $56,676.63 |
| Risk reduction · Expected avoided loss · included | 6.5% | 93.5% | $77,137.50 | $62,828.00 |

### Driver contribution · horizon totals

03

# Time savings · show the math

Capacity value · haircut 6.5% · confidence 93.5%

0 measured · 1 documented · 0 benchmark · 0 estimate · 0 assumption

| Ref | Step / formula / source | Year 1 | Year 2 | Year 3 | Horizon total |
|---|---|---|---|---|---|
| A1 | 1. Gross annual value staff x hours saved per week x weeks x loaded rate (10 x 5 x 40 x $50) Source: user; tier: documented | $100,000.00 | $100,000.00 | $100,000.00 | $300,000.00 |
| A2 | 2. After adoption (50% / 80% / 90%) gross x adoption by year Source: calculated; tier: calculated | $50,000.00 | $80,000.00 | $90,000.00 | $220,000.00 |
| A3 | 3. After recapture (capacity value only) after adoption x recapture 50% Source: default; tier: benchmark | $25,000.00 | $40,000.00 | $45,000.00 | $110,000.00 |
| A4 | 4. After evidence haircut (6.5%) previous step x (1 - haircut) Source: calculated; tier: calculated | $23,375.00 | $37,400.00 | $42,075.00 | $102,850.00 |

### What makes this vary

Freed time may not be turned into other work (recapture is a default of 50%) Adoption ramp depends on training

How measured: Time study on the support queue before and after launch, reviewed monthly by the ops lead

Freed time is capacity value, not cash [A4].

03

# Data quality · show the math

Hard savings · haircut 4.5% · confidence 95.5%

0 measured · 3 documented · 0 benchmark · 0 estimate · 0 assumption

| Ref | Step / formula / source | Year 1 | Year 2 | Year 3 | Horizon total |
|---|---|---|---|---|---|
| B1 | 1. Gross annual value (errors before - errors after) x cost per error ((72 - 24) x $690) Source: user; tier: documented | $33,120.00 | $33,120.00 | $33,120.00 | $99,360.00 |
| B2 | 2. After adoption (50% / 80% / 90%) gross x adoption by year Source: calculated; tier: calculated | $16,560.00 | $26,496.00 | $29,808.00 | $72,864.00 |
| B3 | 3. After evidence haircut (4.5%) previous step x (1 - haircut) Source: calculated; tier: calculated | $15,814.80 | $25,303.68 | $28,466.64 | $69,585.12 |

### What makes this vary

Error baseline covers one year only Cost per error includes only expedite freight

How measured: Monthly expedite-freight invoices compared with the 12-month baseline

Fewer expedited shipments are hard savings [B3].

03

# Risk reduction · show the math

Expected avoided loss · haircut 6.5% · confidence 93.5%

0 measured · 3 documented · 0 benchmark · 0 estimate · 0 assumption

| Ref | Step / formula / source | Year 1 | Year 2 | Year 3 | Horizon total |
|---|---|---|---|---|---|
| C1 | 1. Gross annual value (ALE before - ALE after), ALE = events per year x loss per event ((3 - 1.5) x $25,000) Source: user; tier: documented | $37,500.00 | $37,500.00 | $37,500.00 | $112,500.00 |
| C2 | 2. After adoption (50% / 80% / 90%) gross x adoption by year Source: calculated; tier: calculated | $18,750.00 | $30,000.00 | $33,750.00 | $82,500.00 |
| C3 | 3. After evidence haircut (6.5%) previous step x (1 - haircut) Source: calculated; tier: calculated | $17,531.25 | $28,050.00 | $31,556.25 | $77,137.50 |

### What makes this vary

Outage frequency is a small sample Loss per event excludes reputational cost

How measured: Incident log: sustained outages per year, reviewed quarterly

Halving outage frequency removes expected loss [C3].

## 04 Costs and uplift

PV $75,318.56 · delay 0 months

| Ref / cost | Basis | Base | Uplift | Adjusted |
|---|---|---|---|---|
| D1 · Vendor license, integration and migration (one-time) | quoted | $30,000.00 | 15.0% | $34,500.00 |
| D2 · Training and change management (one-time) | estimated | $5,000.00 | 30.0% | $6,500.00 |
| D3 · Annual run cost | quoted | $12,000.00 | 15.0% | $13,800.00 |

## 05 Unquantified benefits and flexibility

| Benefit | Reason | What to add |
|---|---|---|
| Decision speed | Not selected as a goal for this project | Add the decision speed goal and a dated example decision |

The data pipeline built here could be reused for a second team at lower cost.

06

# Financial summary

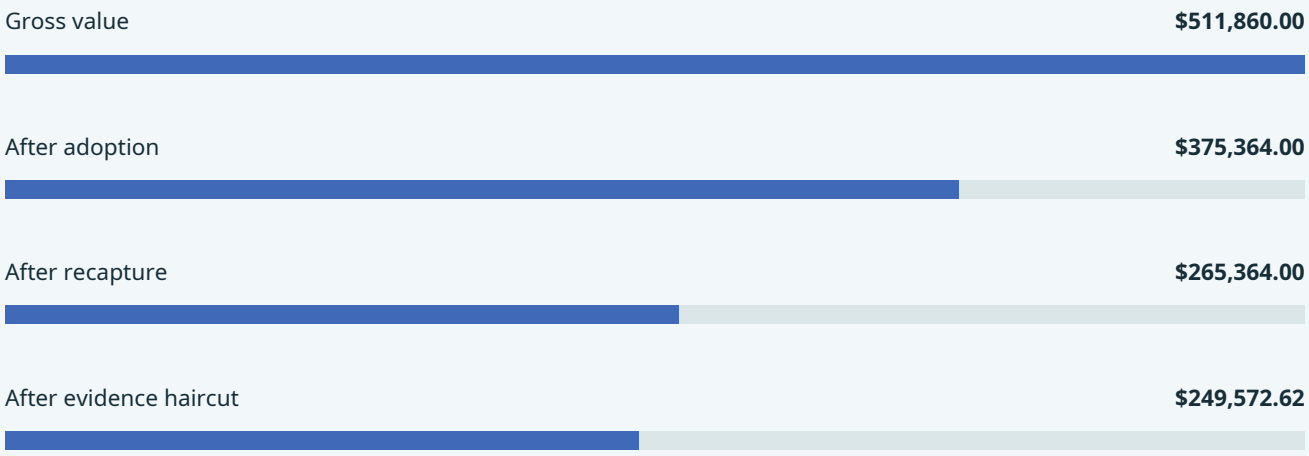
Undiscounted flows; net and cumulative are capacity-inclusive in this sample.

| Period | Hard benefit | Capacity / avoided loss | Cost | Net inclusive | Cumulative inclusive |
|---|---|---|---|---|---|
| Time zero | $0.00 | $0.00 | $41,000.00 | -$41,000.00 | -$41,000.00 |
| Year 1 | $15,814.80 | $40,906.25 | $13,800.00 | $42,921.05 | $1,921.05 |
| Year 2 | $25,303.68 | $65,450.00 | $13,800.00 | $76,953.68 | $78,874.73 |
| Year 3 | $28,466.64 | $73,631.25 | $13,800.00 | $88,297.89 | $167,172.62 |

### Financial basis

Expected avoided loss is distinct from hard savings. The sample's benefits_capacity series includes both time capacity and avoided loss. No hard-only net or cumulative series has been derived.

### Adjustment stages · supplied horizon levels



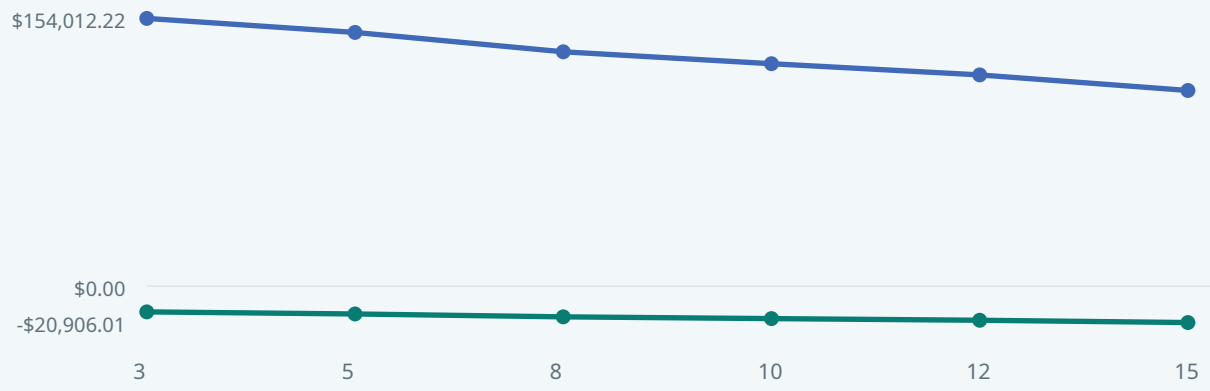

These are successive levels, not floating reduction bars. The engine will need to provide explicit starts and ends for a true waterfall.

## 07 Scenarios and sensitivity

| Scenario / assumptions | NPV hard | NPV inclusive | ROI · PV-based inclusive |
|---|---|---|---|
| conservative: Haircuts at the top of each band, adoption x0.85, recapture 40% | -$29,917.91 | $76,318.52 | 101.3% |
| expected: Headline scenario: stated inputs and computed haircuts | -$18,641.93 | $127,956.73 | 169.9% |
| optimistic: Haircuts at the bottom of each band, adoption x1.10 (capped at 99%), recapture 60% | -$10,036.57 | $182,143.18 | 241.8% |

### NPV sensitivity · discount rate

Teal: hard savings · blue: capacity-inclusive




| Discount rate (%) | NPV hard | NPV capacity-inclusive |
|---|---|---|
| 3 | -$14,778.51 | $154,012.22 |
| 5 | -$15,977.37 | $145,951.48 |
| 8 | -$17,629.04 | $134,810.80 |
| 10 | -$18,641.93 | $127,956.73 |
| 12 | -$19,590.98 | $121,518.06 |
| 15 | -$20,906.01 | $112,567.99 |

## 08 Risks of doing nothing vs proceeding

| Decision | Risks |
|---|---|
| Doing nothing | Expedite-freight spend and outage exposure continue at today's rates. |
| Proceeding | Adoption may ramp more slowly than 50% / 80% / 90%. |
| Proceeding | Vendor costs are quoted, but internal training time is estimated. |

## 09 Measurement plan

| Driver / metric | Baseline / source | Owner | Frequency |
|---|---|---|---|
| Time savings: Time study before and after launch | 30 minutes per ticket; Ops time sample, Q3 | Ops lead | monthly |
| Data quality: Expedite invoices vs 12-month baseline | 72 expedited shipments per year; Finance export | Finance analyst | monthly |
| Risk reduction: Sustained outages per year | 3 per year; Incident log | IT manager | quarterly |

## 10 Appendix and audit trail

| Field | Recorded value |
|---|---|
| Run | run_sample_released |
| Mode | stub |
| Contract | 1 |
| Formula | 0.1 |
| Settings | s1 |
| Generated | 2026-11-05T12:00:00Z |
| Evidence tiers | 0 measured · 7 documented · 0 benchmark · 0 estimate · 0 assumption |
| Narrative references | References such as A4, B3 and D1–D3 refer to the step and cost tables in this report. |

| Engine integrity check | Status |
|---|---|
| bounds | Passed |
| versions recorded | Passed |
| every number traces to a step | Passed |
| driver totals reconcile to benefit total | Passed |
| cash flows reproduce NPV | Passed |
| no unresolved double count | Passed |
| no missing essential cost | Passed |

No warnings were supplied in this sample.

No overrides were supplied in this sample.